Static Analysis of Frames & Machines

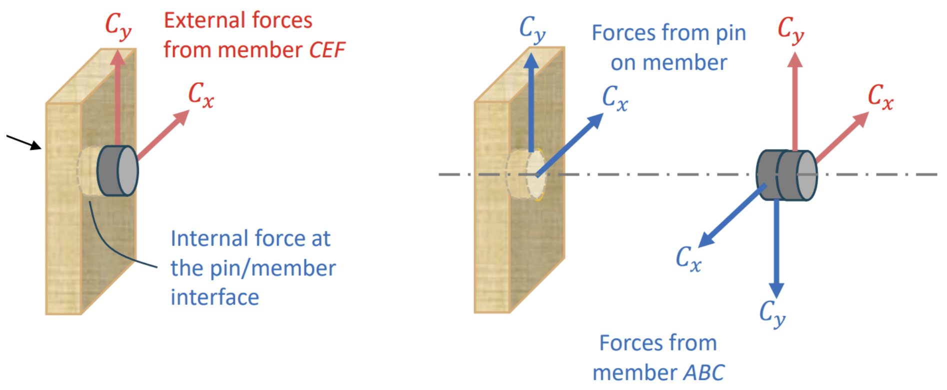
Frames and machines are structures with at least one multi-force member, where frames are meant to be rigid structures while machines are meant to move, e.g. to amplify a force.

The typical method of analysis is to break up the structure into individual rigid bodies, and consider their FBDs, which could be individual members, the whole structure, or sub-structures.

One of the concerns is how to deal with pin joints in the middle of members, as that was not covered in analysis of trusses. For trusses, we ignored the pin and only considered the reaction force of the pin on the member, but can we do the same for multi-force bodies?

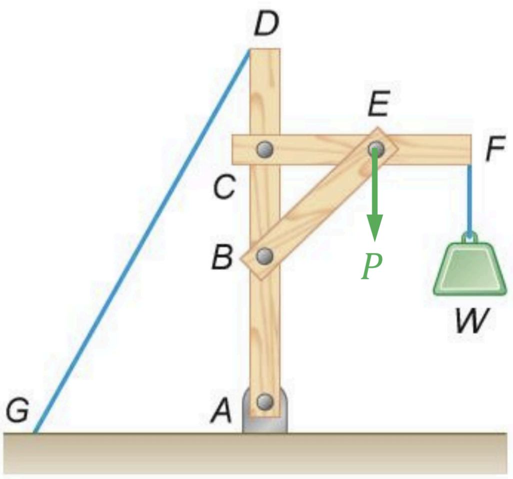
Firstly, for all cases we cannot ignore the pin outright, and must include in one of the FBDs.

In the case when no external forces act on the pin, it does not matter which FBD we attach them to, as the external force is equal to the internal force:



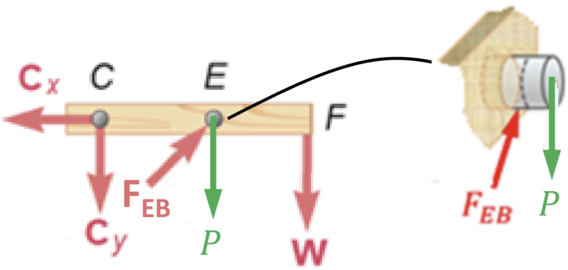
However, when there is an external force on the pin, the reaction forces on the two different members would no longer be equal.

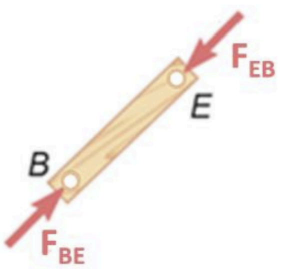
Say we have an external force P here:



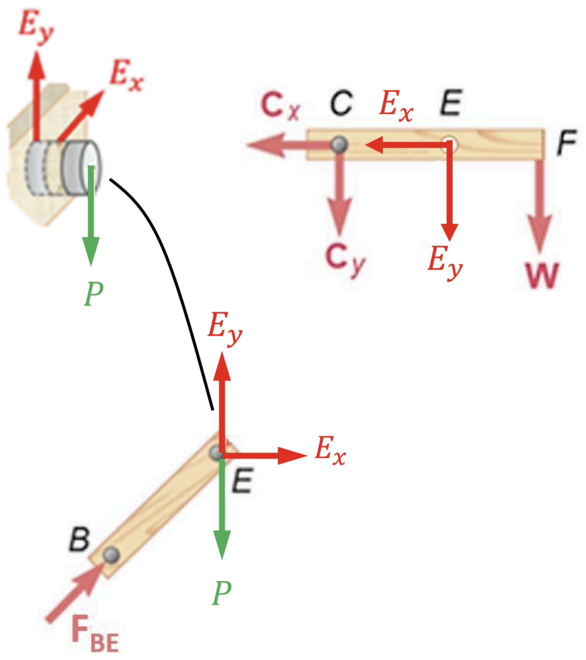
If we let it be attached to CEF, we need to consider the other forces acting on the pin.

In this case, BE acts on the pin E alongside P, pushing against it as BE undergoes (assumed) tension:





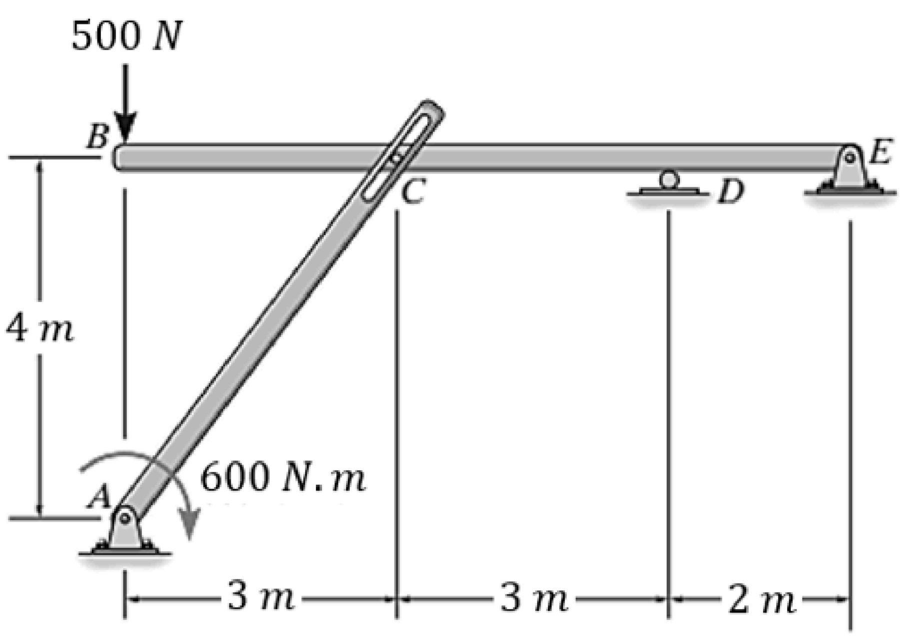
However, if we let the pin be attached to BE instead, the action reaction pair of BE acting on E and E acting on BE are cancelled out, however the action reaction pair of E on CEF is exposed:



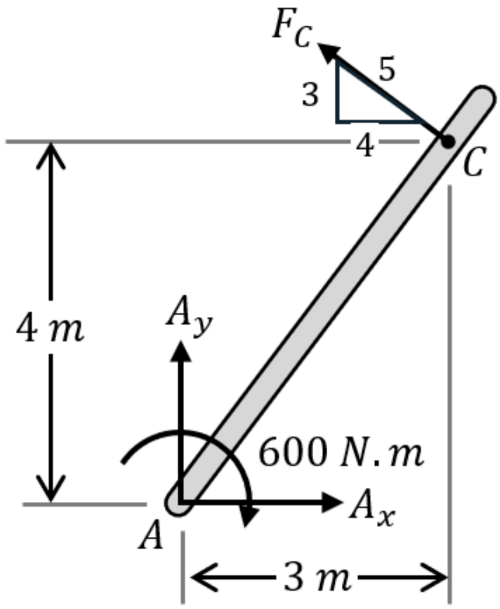
Depending on the question, either FBDs may be used to solve for the specified unknowns.

Another thing to note is that the direction of the action reaction pair of different FBDs must be accounted for, e.g. Ex on CEF is opposite Ex on BE.

Say we had to find the reaction forces at A and E in the structure shown:



We cannot solve it directly as there are 5 unknowns present. Instead, split it into the individual FBDs. Then, consider AC:



There are only 3 unknowns, thus we can use static analysis.

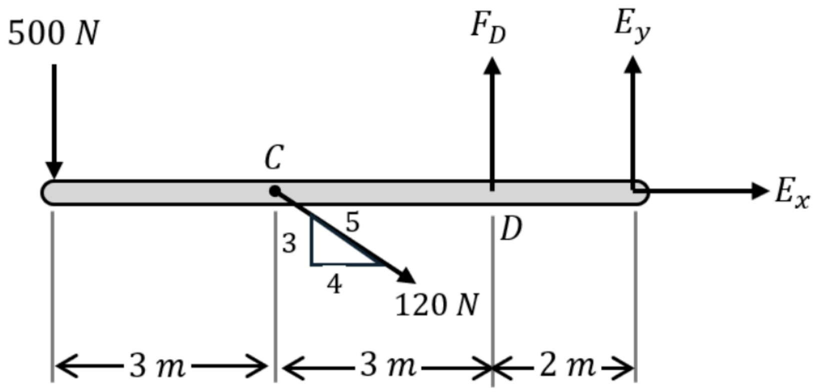
$$\sum M_A = 0$$
$$F_C(\sqrt{3^2+4^2}) - 600 = 0$$
$$F_C = \frac{600}{5} = 120N$$

From here, we could solve it using $\sum F_x$ and $\sum F_y$, but since only two forces are present, the forces should balance each other out for translational equilibrium.

Hence,

$$R_A = F_C$$
$$= 120N\ (\angle - 36.86^\circ)$$

From here, either the full assembly or BCDE can be used to solve R_E, so if we use BCDE:



$$\sum M_D = 0$$
$$2E_y + 500(6) + \frac{3}{5}(120)(3) = 0$$
$$E_y = -1608N$$

Then,

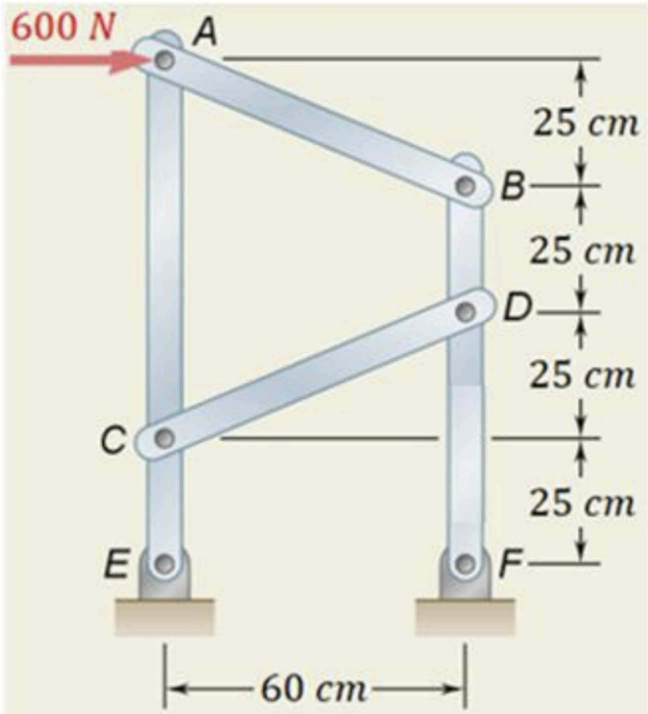
$$\sum F_x = 0$$
$$E_x + \frac{4}{5}(120) = 0$$
$$E_x = -96N$$

Hence the magnitude is

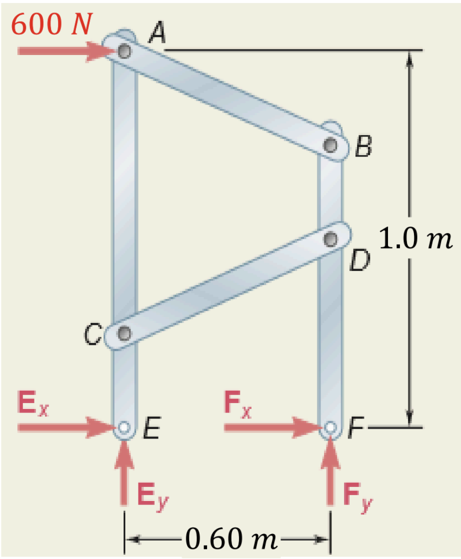
$$|E| = \sqrt{1608^2 + 96^2} = 1611N$$

For that question, no external forces were applied on the pins, however, consider the question:

Given the structure, find the forces in members AB and CD.



We can first look at the whole assembly:

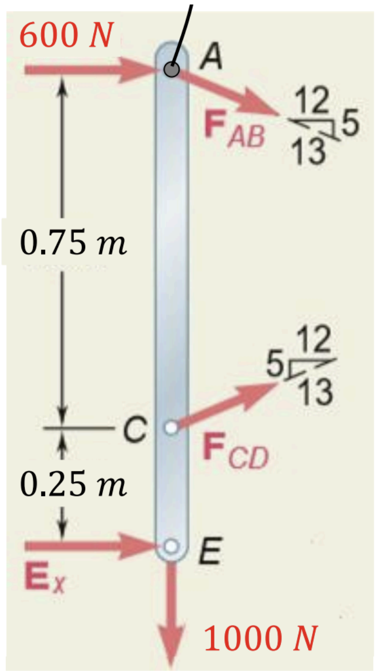


And find that

$$\sum M_E = 0$$
$$F_y(0.6) - 600(1) = 0$$
$$F_y = \frac{600}{0.6} = 1000N$$

$$\sum F_y = 0$$
$$F_y + E_y = 0$$
$$E_y = -1000N$$

Now, attatch the pin to ACE, to expose the force AB acting on the pin, then consider the member ACE:



then,

$$\sum M_E = -600(1) - \frac{12}{13}F_{AB}(1) - \frac{12}{13}F_{CD}(0.25) = 0$$

$$\sum F_y = -\frac{5}{13}F_{AB} + \frac{5}{13}F_{CD} - 1000 = 0$$

which forms two simultaneous equations which can be solved to get

$$F_{AB} = -1040N, F_{CD} = 1560N$$

Had we attached the pin to AB, we have to use AB to solve for F_{AB}, and can only for solve it seperately from F_{CD}.